



Predicting Unemployment Using Economic Indicators: A Machine Learning Approach

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Introduction

One of the critical indicators of a country's economic health and labour market stability is unemployment. It represents the share of the labour force that is willing and able to work but is without a job. Elevated unemployment levels may indicate economic inefficiencies, lower productivity, and social issues such as income inequality and a fall in living standards. Hence, economists and policymakers are keen to understand what factors influence unemployment.

Various macroeconomic indicators have the potential to influence unemployment rates. Economic expansion leads to job generation, inflation is a measure of price stability and economic conditions, labour force participation rate is the percentage of people who are either employed or actively looking for work, and population growth is the demographic factor that changes the labour supply. Nonetheless, the nature of these relationships with unemployment is often intricate and different for each country.

The research takes advantage of the World Bank's World Development Indicators dataset to study unemployment patterns in a group of countries in the period 1991-2024. Through exploratory data analysis and machine learning techniques, the paper seeks to uncover the economic factors that have the greatest impact on unemployment and also assesses the predictive power of these variables on unemployment levels.

Problem Statement and Objectives

Labour market results, including unemployment, greatly differ by country because of unequal economic growth, labour market structures, and demographics. Although changes in labour market are usually linked with the GDP growth, inflation, labour force participation rate (LFPR), and population growth, it is still unclear which factors have the highest impact on unemployment.

This study primarily intends to explore the link between unemployment and some macroeconomic indicators. In particular, it decides the extent to which these indicators shape unemployment trends and assesses the performance of machine learning models in forecasting unemployment figures based on these input variables.

Dataset and Variables

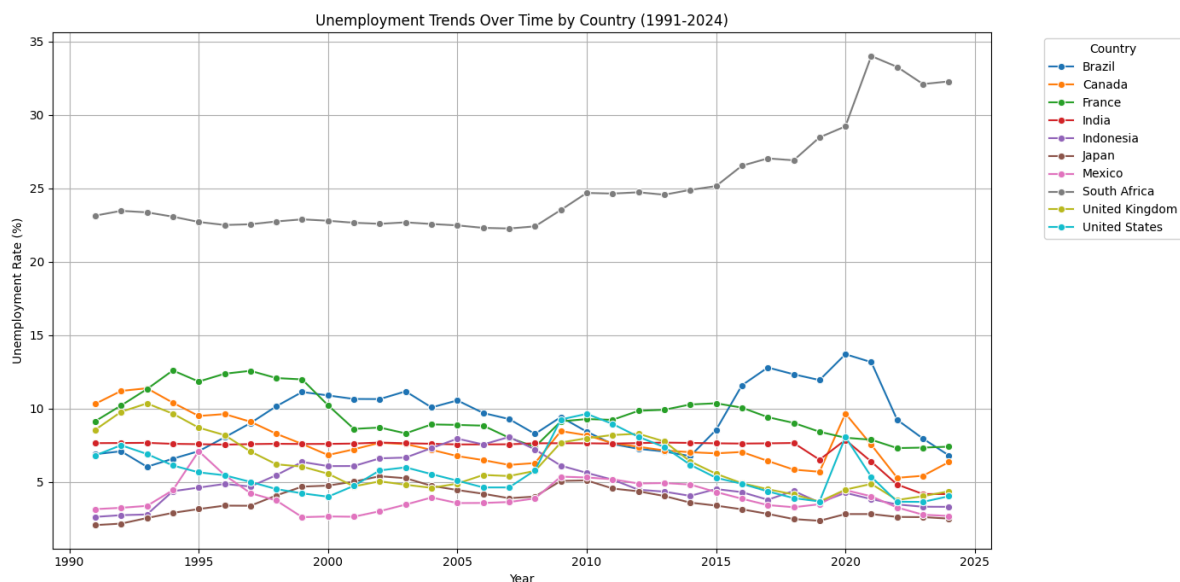
The data for this investigation were sourced from the World Bank World Development Indicators database. The study time frame is 1991 to 2024, and the countries whose data were included are the United States, Japan, France, the United Kingdom, Canada, India, Brazil, Mexico, Indonesia, and South Africa.

This research utilises one dependent variable and four independent variables. The dependent variable is the unemployment rate (% of total labour force). The independent variables are GDP growth, inflation, labour force participation rate (LFPR), and population growth. These variables embody major economic and demographic factors that could have an impact on labour market situations.

Methodology

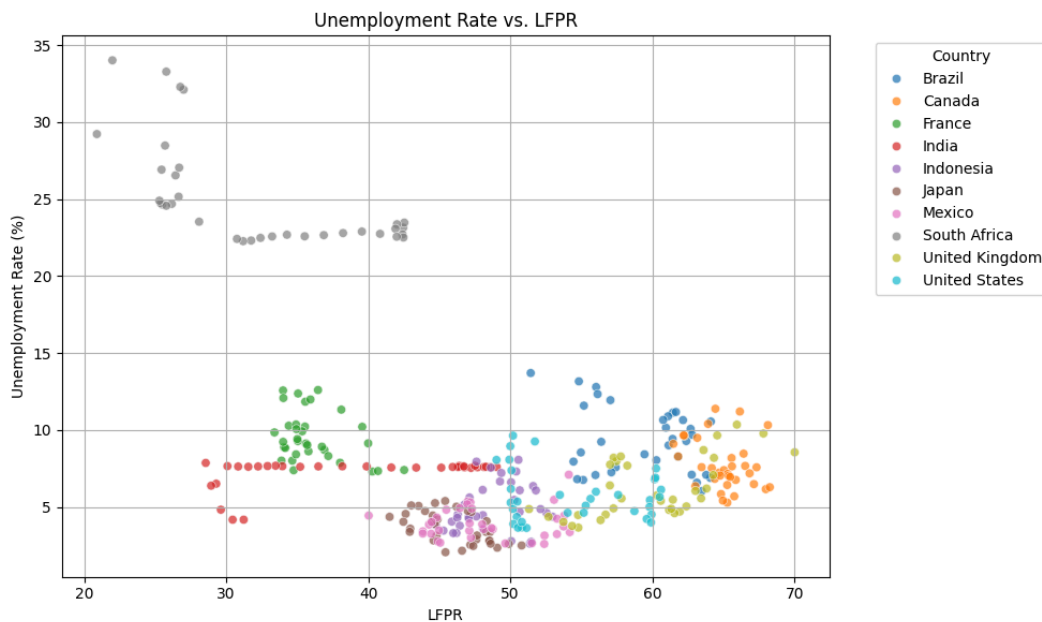
We started by getting the data ready, then went on to explore the data, and finally, we did some machine learning modelling.

Initially, the datasets for each indicator were gathered and merged by country and year to form one dataset. After that, the data was cleaned by the removal of missing values and the data was checked for consistent formatting. Then, exploratory data analysis (EDA) was carried out to find out the data patterns and how unemployment is related to the call of the economic indicators.

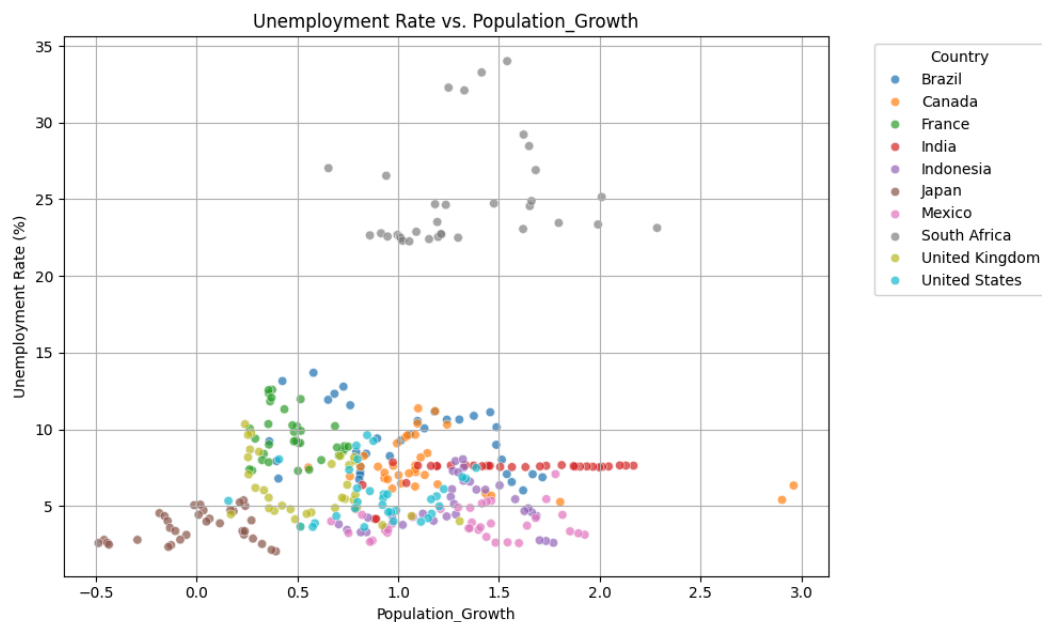


This graph shows how unemployment levels have evolved across different countries over time. The trends highlight significant differences between economies, with some countries maintaining relatively stable unemployment while others experience higher and more volatile rates.

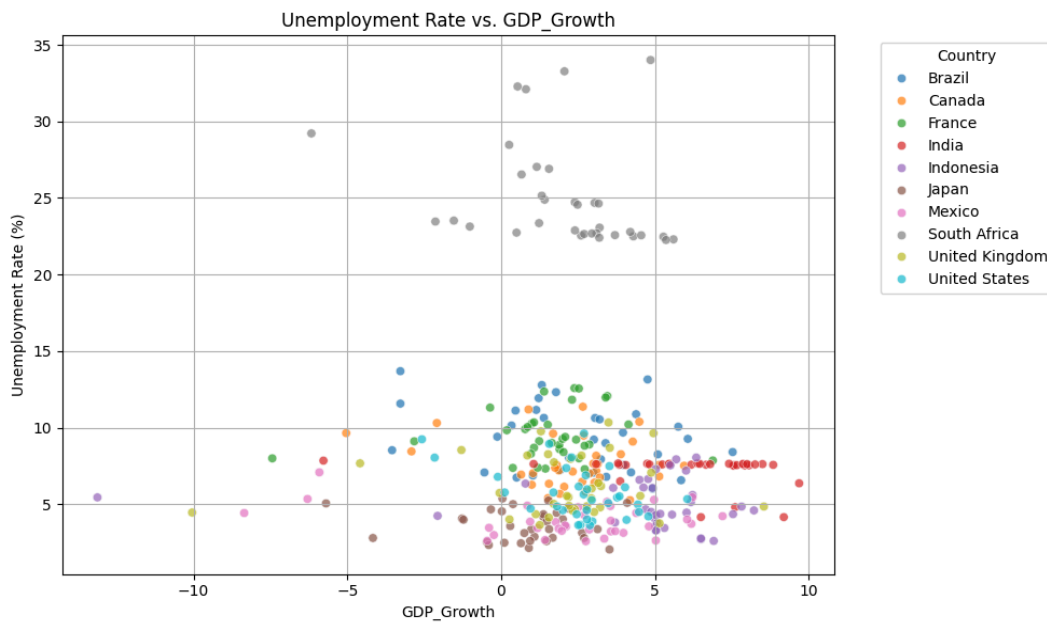
Scatter plots were also used to explore relationships between unemployment and individual economic indicators.



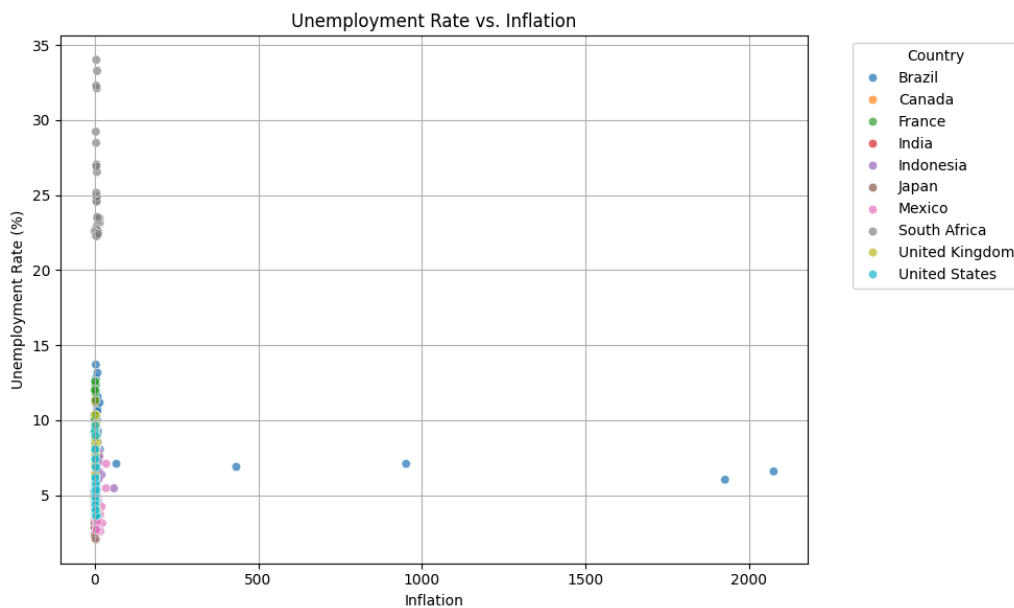
The scatter plot suggests that labour force participation plays an important role in unemployment patterns, with higher participation levels often associated with lower unemployment.



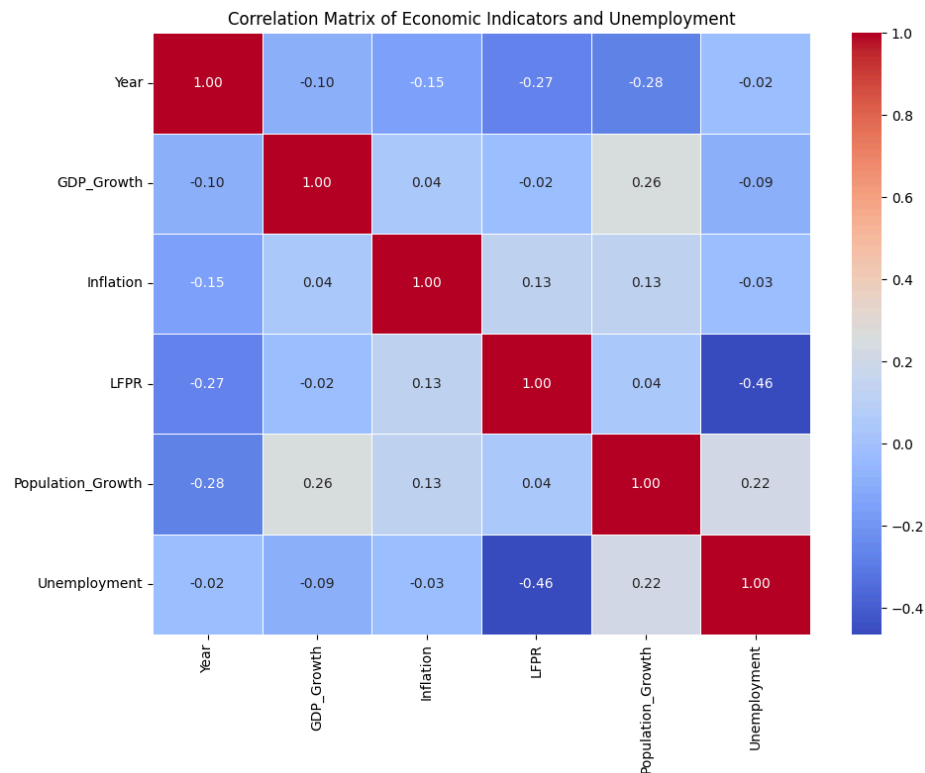
This visualisation shows that population growth alone does not strongly determine unemployment levels, indicating that demographic changes interact with other economic factors.



The relationship between GDP growth and unemployment reflects the economic principle that stronger economic growth generally leads to greater employment opportunities.



The inflation plot indicates a relatively weak direct relationship between inflation and unemployment within the dataset.



The correlation heatmap summarises the relationships between all variables and highlights which indicators show stronger associations with unemployment.

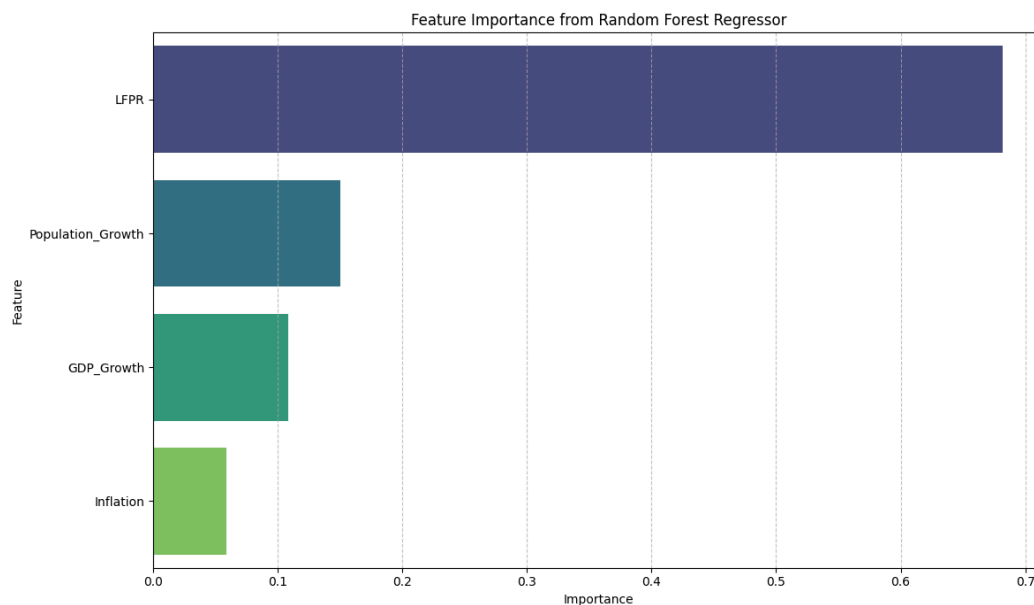
Machine Learning Results

Several machine learning models were applied to analyse the dataset, including Decision Tree Regressor, Random Forest Regressor, K-Nearest Neighbours Regressor, and Support Vector Regressor. The models were evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score.

Model Evaluation Results:					
	Model	MSE	RMSE	R-squared	
0	Decision Tree Regressor	7.332270	2.707816	0.583747	
1	Random Forest Regressor	5.172043	2.274213	0.706383	
2	K-Nearest Neighbors Regressor	7.188241	2.681090	0.591923	
3	Support Vector Regressor	16.471743	4.058540	0.064898	

The model evaluation results show that the Random Forest Regressor performed the best among the tested models, achieving the lowest Mean Squared Error ($MSE = 5.17$) and Root Mean Squared Error ($RMSE = 2.27$), along with the highest R^2 score of 0.71. In comparison, the Decision Tree Regressor achieved an R^2 score of 0.58, while the K-Nearest Neighbours model produced an R^2 score of 0.59. The Support Vector Regressor performed poorly with an R^2 score of 0.06, indicating that it was not well-suited for this dataset. This suggests that the Random Forest model was most effective at capturing the relationships between the economic indicators and unemployment.

To better understand the influence of each variable on unemployment predictions, feature importance analysis was conducted using the Random Forest model.



Feature importance analysis using the Random Forest model reveals that Labour Force Participation Rate (LFPR) is the most influential variable, accounting for approximately 67% of the model's predictive importance. Population growth accounts for about 15%, while GDP growth accounts for about 11% of the model's explanatory power. Inflation shows the lowest importance (around 6%), suggesting it plays a relatively weak role in predicting unemployment in the dataset.

Key Findings and Recommendations

Unemployment patterns differ greatly from one country to another because of the different economic structures and labour market dynamics of each. Labour force participation is the variable most strongly related to unemployment rates, according to the research, so policies that promote workforce participation could have a big impact on labour market results.

At the same time, economic growth is a very important factor influencing unemployment trends and underlines the need for a continuously and widely growing economy that will create more jobs. Growth of the population has a less influential role, which suggests that, over time, labour markets could be affected by demographic pressures.

It is our opinion that authorities should put their efforts into encouraging workforce participation through suitable skills development, training and provision of flexible work arrangements. Besides that, by attracting investments and working towards becoming a more innovative economy, it is possible to achieve a higher standard of employment. Lastly, there is also the possibility of the governments using techniques such as machine learning for labour market monitoring and policy making, which would enable identifying conditions and changes better and thus getting the most useful policies.